

Reproducible Lit Review

A literature review is finite only when it serves a decision. This skill makes you name the decision first, then walks six stages that end in experiments with falsifiers — not a pile of PDFs.

github.com/Goutham-Eda/reproducible-lit-review MIT 26 files Python 3.9+, stdlib only
Installed at `~/.claude/skills/lit-review` — a clone, so edits there are commits

Why it exists

Most review tooling optimises for finding more papers. The failures that actually cost time are structural — and they share one shape: **a channel that cannot carry the answer.**

FIVE OF THE SIX FAILURE MODES, AND WHAT WAS MISSING

THE FAILURE	WHAT WAS MISSING
Ranking rule has nothing to rank on	no tier to rank in
Citation graph can't reach a literature	no edge to traverse
Extraction tool mis-keys a row	no key to join on
No column for an unanticipated answer	nowhere to write it
Metric reports zero for every condition	no range to move in

After the fact you cannot distinguish *no signal* from *nowhere to put it* — which is why each one is a check that runs **before** the stage it protects.

The six stages

Each stage writes a file the next one reads, so the whole review is a folder someone else can re-run. **That file contract is the reproducibility claim.**

WRITES `clusters.csv`

2 Seeds — papers you already trust

WRITES `seeds.csv`

3 Harvest — citation graph, both directions, then a name search

WRITES `candidates.csv` `named.csv` `overlap.md`

4 Shortlist — screened, with provenance on every row

WRITES `shortlist.csv` `funnel.md`

5 Extraction — quote-grounded columns with escape hatches

WRITES `shortlist_extracted.csv`

6 Experiments — findings that can be wrong

WRITES `experiments.md`

Stage 1 is the irreducibly human one. Clusters come from a practitioner describing the problem in their own words — a transcript, a protocol, an expert's tour of their data — not from a published taxonomy (organised for teaching) and not from your own hypotheses (then the filing system agrees with you by construction). Pull out the recurring *questions*, not topics: a question has an answer, so a paper either answers it or doesn't, and that is screenable.

Stage 6 is also human. No script can pick a falsifier.

The guards, concretely

Co-citation ranking collapses when seeds share no references

`harvest.py` writes `overlap.md` — the pairwise matrix, and which ranking rule is available — before you write one

A graph never finds work that cites nothing you seeded

`rank.py` exits non-zero if `named.csv` is absent; `--allow-graph-only` overrides and records the gap in `funnel.md`

Extraction tools mis-key rows, silently

`fetch_pdfs.py` writes a manifest keyed on the filename you chose; `table.py join` uses it first, then exact title, then fuzzy ≥ 0.92 , reporting every fuzzy and unmatched row

“Not found in my corpus” written up as “unclaimed in the field”

`table.py check` greps every cell for *unclaimed, nobody has, first to, no prior work*

No column for an answer you didn't anticipate

`table.py check` fails if `suggests_new_question` is missing — and fails again if it exists and is empty, since an empty column is the same defect wearing a header

A metric that only ever reports zero

The register requires a floor check on a random or trivial policy before a metric becomes primary

The scripts

Stdlib-only Python 3.9+, no install step. OpenAlex is primary — no key, both citation directions. arXiv is the recency channel, because it carries preprints within a day.

SCRIPTS/

SCRIPT	JOB
<code>oa.py</code>	OpenAlex + arXiv client, disk cache, backoff, and the pipeline's single paper schema
<code>harvest.py</code>	Backward + forward hops, <code>seed_overlap</code> , the overlap diagnostic
<code>name_search.py</code>	Direct name search — a required separate step, because the separation is the point
<code>rank.py</code>	The funnel → <code>shortlist.csv</code> + <code>funnel.md</code>

SCRIPT	JOB
<code>fetch_pdfs.py</code>	PDFs + <code>manifest.csv</code> , the join key
<code>table.py</code>	<code>skeleton</code> <code>join</code> <code>check</code>

The funnel's rules, each with a scar behind it

- **≥ 2 cluster-keyword hits.** One is noise — “Long Short-Term Memory” matches a memory cluster.
- **A fame ceiling.** Above ~25k citations a paper is infrastructure or a textbook, not a row you extract a mechanism from.
- **A recency floor,** waived by co-citation.
- **Per-seed depth,** so one deep literature can't swamp the rest.
- **Then a cluster quota.**

Never rank on citation count — done for real, it put a plotting library at the top of a mechanism cluster and pushed every mechanism paper out. Hand judgement in `overrides.csv` bypasses the guards and survives every re-run.

Two design commitments

Escape hatches are load-bearing

Every extraction column carries an explicit permitted non-answer — “`not reported`”, “`no evidence`”, “`none - <type>`” — written into the question itself. Without it, gaps get filled with plausible inference and the column answering your actual question becomes confident fiction. A high escape-hatch rate is the *good* outcome.

Nothing paid sits in the critical path

Every stage degrades: the extraction table is hand-fillable from abstracts already in `shortlist.csv`. The review gets thinner; it does not stop.

Two validation runs

Both on questions unrelated to the project the method came from, and both six seeds. The second was run under a borrowed stopping rule — 15–20 sources, reviews and handbook

chapters only — which is what exposed the type filter.

Sleep staging

Do sleep-staging models generalise across devices?

Pooled candidates	535
Passed screening	208 39%
Shortlisted	38 7%
Extraction filled	14 3%

Scaled to this run's pool of 535.

Cross-domain origins

Do the fields where the cluster questions were first posed offer a mechanism, an argument, or background only?

Pooled candidates	611
Passed screening	31 5%
Shortlisted	16 3%
PDFs obtained	1 0.2%

Scaled to this run's pool of 611.

What the sleep run showed

- 13 of 38 shortlisted papers were reachable only by name search — a third of the shortlist, invisible to both graph directions, including the two that most directly answered the question.
- The consumer-device validation literature and the deep-learning staging literature share zero citations. That fell out of the overlap matrix in stage 3, before a paper was read — and it is the answer to the question.
- 10 of 14 rows answered “not reported” on the one metric the decision needed. The escape hatch is the only reason that finding was visible.

- **Join tested both ways:** title-only matched 12/14 with two rows collided on one key; filename-via-manifest matched 14/14.
- **PDF coverage was 28.6%**, against 96.6% on an arXiv-heavy shortlist — clinical publishers return 403 to any non-browser client. Reported, not hidden.

It also found a defect in its own ranking: best-cluster-wins starved the “does it transfer” cluster to 2 of 8 and called it a thin literature. It wasn't — any EEG abstract is dense in EEG words, so the sparser-vocabulary cluster loses. Papers now count toward any cluster they hit. **A keyword-derived “thin literature” is a claim about your keyword lists until you've checked it.**

What the cross-domain run showed

Gibson's affordances, Gopnik and Spelke, Ashby's requisite variety, Powers' perceptual control.

TWO DEFECTS, BOTH FIXED AND RE-RUN

DEFECT	CONSEQUENCE	FIX
Forward hops sorted by publication date	For a 1979 classic, returns last month's most obscure applications — football tactics, dating-app engagement — rather than the influential citing work	Sorted by citations, with <code>--forward-sort</code> to override
No document-type filter, while the stopping rule admits only reviews and chapters	The first run returned zero of either in 18 rows	<code>rank.py --types review,book-chapter,book</code> screens on the OpenAlex <code>type</code> field — 362 article rows excluded on the re-run — and <code>funnel.md</code> reports how many rows carry no type and so escaped the screen

Two structural findings

- **For pre-1990 books the backward channel is dead.** Gibson returned 0 references, Ashby 3, Powers 6 — OpenAlex holds no reference lists for books. The exact mirror of a corpus whose seeds are too new.
- **14 of 15 seed pairs share zero references**, max overlap 2. Four separate origins for four cluster questions, unconnected.

The substantive result was a defensible negative. Cluster D's *review* literature turned out to be Information Systems affordance theory rather than ecological psychology, so no source in that corpus engages Gibson's premise where it bears on the controllability hypothesis. Stated as “not found in this corpus, which under-samples ecological psychology” — the wording the skill enforces.

What it does not do

It is **not PRISMA**, and says so: the method is citation-graph snowballing plus a direct name search plus hand triage. It does not read the papers, derive your clusters, or choose your falsifiers.

[references/worked-example.md](#) closes with the ten points where the real run needed human judgement — rejecting four keyword-passing papers that were about sleep but not about staging models; marking the most co-cited item in the pool as a dataset with no mechanism to extract; telling “thin literature” apart from “starved by my own scoring rule”. That list is the honest scope of the tooling.

[reproducible-lit-review](#) · MIT · OpenAlex + arXiv, no API key